

operating system

An operating system is the software that controls the overall operation of a computer.

操作系统是控制计算机整体运行的软件。

- It provides the means to store and retrieve files. 存储和检索文件
- It provides an interface for users to request program execution. 用户请求程序执行的接口
- It provides the environment necessary to execute programs. 程序运行所需的环境

3.1 The History of Operating Systems

Job Each program execution was called a job.

Each job was processed as an isolated activity.

Preparation and cleanup were required before and after each job.

batch processing (批处理) Batch processing is the execution of jobs collected into a batch.

Jobs are executed without user interaction.

The operating system reads and executes jobs one at a time.

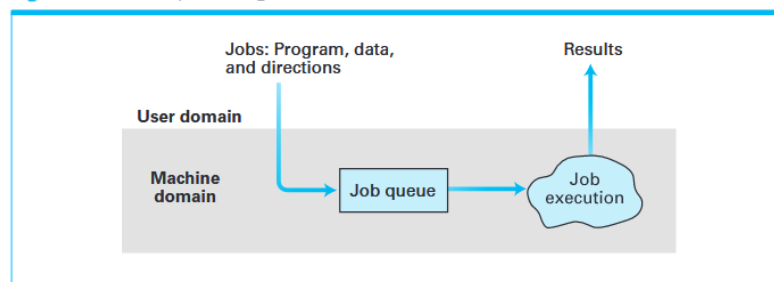
job queue Jobs waiting for execution are stored in a job queue.

first-in, first-out (FIFO)

Real systems do not strictly follow FIFO. Jobs may have priorities.

A high-priority job can bypass(插队) others.

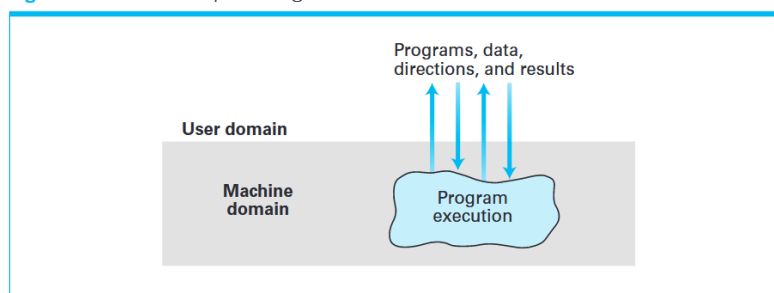
Figure 3.1 Batch processing



interactive processing

Interactive processing allows a program to communicate with the user during execution.

Figure 3.2 Interactive processing



real-time processing

Real-time processing means tasks are completed within strict time constraints.

time-sharing 分时 → 多用户

Time-sharing allows multiple users to use a computer at the same time.

Multiprogramming

Multiprogramming divides CPU time into intervals and switches between jobs.

Multitasking 多任务 → 单用户

Multitasking refers to a single user running multiple tasks simultaneously.

Multiprocessor Systems (多处理器系统)

-load balancing (负载均衡)

Load balancing distributes tasks evenly among processors.

-Scaling (扩展性)

Scaling means dividing a task into subtasks for multiple processors.

embedded systems

Embedded systems are computers dedicated to specific tasks.

1. Identify examples of queues. In each case, indicate any situations that violate the FIFO structure.

(1) People waiting in line at a ticket counter

排队买票的人

→ FIFO: First person in line is served first

→ 先排队的人先被服务

Violation: VIP customers or emergency cases may be served earlier

违反: VIP 或紧急情况可以插队

(2) Jobs in a job queue in batch processing systems

批处理系统中的作业队列

→ FIFO: Jobs are executed in order of arrival

→ 按到达顺序执行

Violation: High-priority jobs may bypass others

违反: 高优先级作业可以插队

(3) Printer queue

打印机队列

→ FIFO: Documents are printed in submission order

→ 按提交顺序打印

Violation: Urgent print jobs may be prioritized

违反: 紧急打印任务可能优先

2. Which of the following activities require real-time processing?

a. Printing mailing labels

b. Playing a computer game

c. Displaying numbers on a smartphone screen as they are dialed

d. Executing a program that predicts the state of next year's economy

e. Playing an MP3 recording

b c e

3. What is the difference between embedded systems and PCs?

Embedded systems are computers dedicated to specific tasks.

PCs are general-purpose computers designed to perform a wide range of tasks.v

4. What is the difference between time-sharing and multitasking?

time-sharing 分时 → 多用户

Time-sharing allows multiple users to use a computer at the same time.

Multitasking 多任务 → 单用户

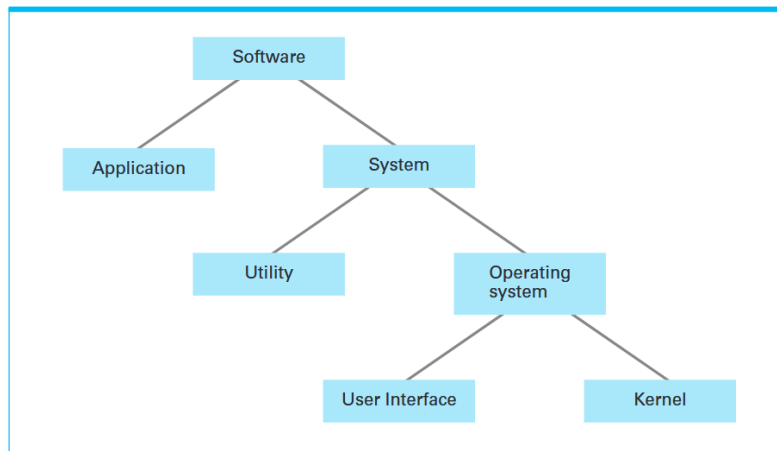
Multitasking refers to a single user running multiple tasks simultaneously.

3.2 Operating System Architecture

A Software Survey

Software Classification

Figure 3.3 Software classification



-application software

It performs tasks specific to the user's needs.

Example: Spreadsheets → 电子表格

Database systems → 数据库系统

Accounting systems → 会计系统

Games → 游戏

-system software

It performs tasks common to all computer systems.

It provides the infrastructure for application software.

--Operating System

It manages hardware and provides an environment for applications

--utility software

It performs tasks not included in the operating system

It extends or enhances the capabilities of the OS

Example: File management utilities → 文件管理工具

Disk formatting tools → 磁盘格式化工具

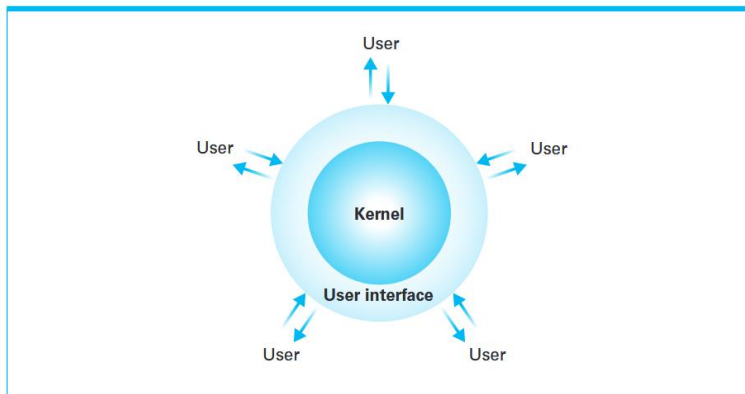
Data compression software → 数据压缩软件

Multimedia software → 多媒体播放软件

Network communication software → 网络通信软件

Components of an Operating System

Figure 3.4 The user interface acts as an intermediary between users and the operating system's kernel



user interface

It allows users to interact with the system

It acts as an intermediary between the user and the kernel

-shells (命令行界面)

They use textual commands via keyboard and monitor

-graphical user interface (GUI) (图形用户界面)

It represents objects as icons on the screen

--window manager (窗口管理器)

It allocates screen space into windows

It tracks which application owns each window

kernel (内核)

It performs the fundamental operations of the system

-file manager

It keeps track of file locations

It controls access permissions

It manages available storage space

--directory or folder (目录/文件夹)

It groups related files together

Directories can contain subdirectories

This creates a hierarchical structure

--directory path

It lists directories separated by slashes

-device drivers (设备驱动程序)

They communicate with controllers or devices

They translate generic requests into device-specific actions

-memory manager (内存管理器)

It allocates memory space to programs

It ensures programs do not interfere with each other

It tracks free and used memory

--paging (分页)

It moves data between main memory and mass storage

--pages (页)

Memory is divided into pages

--virtual memory

虚拟内存：通过分页扩展的“假想内存” It creates the illusion of larger memory

-scheduler (调度器) It selects processes for execution

-dispatcher (分派器) It allocates CPU time to selected tasks

Getting It Started

boot strapping (booting) (引导过程)

→ 引导 (启动)：计算机开机时启动操作系统的过程

启动是指 ROM 中的引导程序将操作系统从外存加载到主存并将控制权交给它的过程

It transfers the operating system into main memory

Problem of Volatile Memory (易失性内存问题)

Main memory is volatile → 主存是易失性的 (断电数据消失)

Data must be reloaded each time the computer is turned on

→ 每次开机都必须重新加载数据

read-only memory (ROM) (只读存储器)

→ 只读存储器：一种非易失性存储

Its contents remain even when power is off

It stores the initial program for startup

boot loader (引导程序)

→ 引导程序：存储在 ROM 中的小程序

It is the first program executed when the computer starts

It loads the operating system into main memory

It then transfers control to the operating system

Booting Process (启动流程)

The CPU starts execution at a predefined address → CPU 从预设地址开始执行

This address contains the boot loader → 该地址存储引导程序

The boot loader loads the OS from storage → 引导程序从外存加载操作系统

Control is then transferred to the operating system → 然后将控制权交给操作系统

Firmware (固件)

Firmware → 固件：介于硬件和软件之间的软件

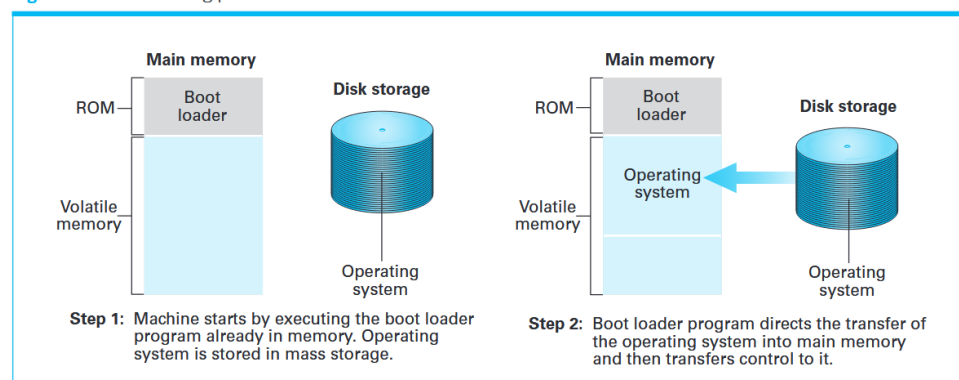
Stored in nonvolatile memory such as FlashROM

It performs basic input/output operations

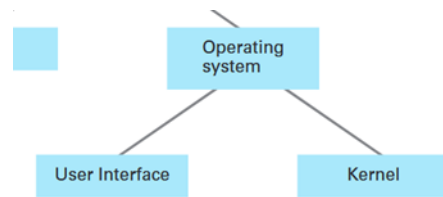
firmware update

update operating systems and boot loaders stored in ROM

Figure 3.5 The booting process



1. List the components of a typical operating system and summarize the role of each in a single phrase.



User interface: It allows users to interact with the system

It acts as an intermediary between the user and the kernel

Kernel: It performs the fundamental operations of the system (file management, memory management, device drivers, process scheduling)

2. What is the difference between application software and utility software?

-application software

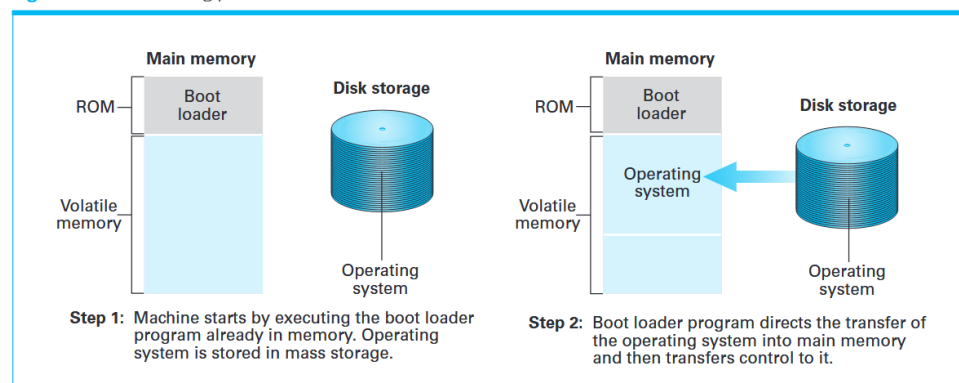
The difference is that application software is designed for specific user tasks, while utility software supports and enhances the operating system.

3. What is virtual memory?

Virtual memory is a technique that uses paging to transfer data between main memory and mass storage, creating the illusion of a larger memory space.

4. Summarize the booting procedure.

Figure 3.5 The booting process



Booting Process (启动流程)

The CPU starts execution at a predefined address → CPU 从预设地址开始执行

This address contains the boot loader → 该地址存储引导程序

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Control is then transferred to the operating system → 然后将控制权交给操作系统

3.3 Coordinating the Machine's Activities

Program vs Process (程序 vs 进程)

Program: a static set of instructions

Process: the activity of executing a program

A program is static, while a process is dynamic

The Concept of a Process

Process the activity of executing a program → 程序执行时的动态过程

It is managed by the operating system

process state

It includes the current position in the program (program counter)

It includes CPU register values

It includes relevant memory contents

Process Administration

Scheduler The scheduler maintains a pool of processes

process table → 进程表: 存储所有进程信息的数据结构

It is maintained by the scheduler in main memory

Each process has an entry in the table

Memory allocation/ Priority/ Process state (ready / waiting)

-ready

The process is ready to use the CPU

-waiting

The process is waiting for an external event

Dispatcher The dispatcher controls execution of processes

multiprogramming (多道程序)

→ 多道程序: 多个进程共享 CPU 执行

It improves efficiency by using CPU time effectively

When one process waits, another executes

→ 一个进程等待时, 另一个进程执行

time slice

→ 时间片: 分配给一个进程的一小段 CPU 时间

It is usually very short (milliseconds)

process switch (context switch)

→ 进程切换: 从一个进程切换到另一个进程

The current process is paused and another is started

interrupt

→ 中断: 打断当前执行的信号

It causes the CPU to stop current execution → 使 CPU 停止当前任务

Control is transferred to the operating system → 控制权转交给操作系统

CPU: 停下当前任务/记录当前位置 (process state) /执行中断处理程序

interrupt handler

Interrupt handler → 中断处理程序: 处理特定中断的程序

It determines how the system responds to an interrupt → 决定系统如何响应中断

Interrupt handler tells the dispatcher what to do

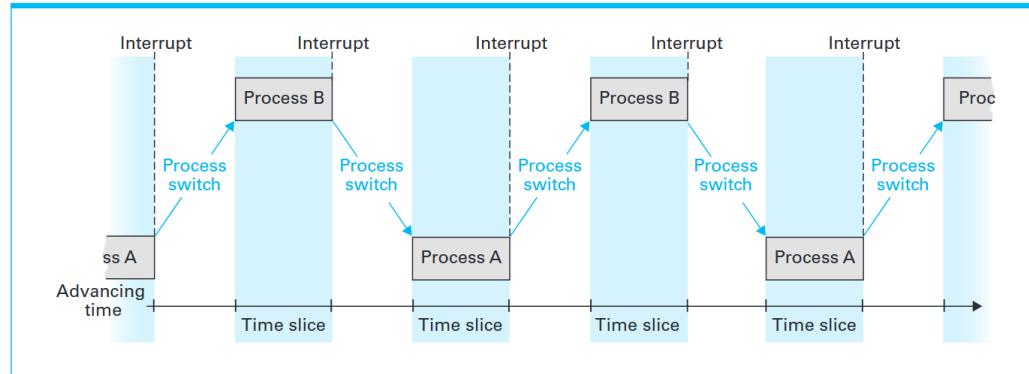
Interrupts

→ 中断系统：用于协调计算机与外部环境

They allow the CPU to respond to events → 使 CPU 能够响应各种事件

Example: Keyboard input/ Mouse actions/ I/O completion/ Power failure

Figure 3.6 Multiprogramming between process A and process B



1. Summarize the difference between a program and a process.

Program is a static set of instructions. Process is the activity of executing a program.

A program is static, while a process is dynamic.

2. Summarize the steps performed by the CPU when an interrupt occurs.

CPU: 停下当前任务/记录当前位置 (process state) /执行中断处理程序

When an interrupt signal is received, the CPU completes its current machine cycle, saves its position in the current process, and begins executing a program called an interrupt handler.

3. In a multiprogramming system, how can high-priority processes be allowed to run faster than others?

In a multiprogramming system, high-priority processes run faster because the dispatcher repeatedly selects them for execution when assigning time slices.

4. If each time slice in a multiprogramming system is 50 milliseconds and each context switch requires at most a microsecond, how many processes can the machine service in a single second?

$$1 \text{ time slice} = 50 \text{ milliseconds} = 0.05 \text{ s}$$

$$1 \text{ context switch} \leq 1 \text{ microsecond} = 0.000001 \text{ s}$$

$$0.05 + 0.000001 \approx 0.050001 \text{ s}$$

$$1 / 0.050001 \approx 19.9996 \approx 20 \text{ processes}$$

5. If each process uses its complete time slice in the machine in question 4, what fraction of the machine's time is spent actually performing processes?

$$\text{time slice} = 50 \text{ ms} = 0.05 \text{ s}$$

$$\text{context switch} = 1 \text{ } \mu\text{s} = 0.000001 \text{ s}$$

$$0.05 / 0.050001 \approx 0.99998$$

What would this fraction be if each process executed an I/O request after only a microsecond of its time slice? $0.01 / 0.01 + 0.01 = 0.5$

3.4 Handling Competition Among Processes

Resource allocation is the task of assigning system resources to processes

Semaphores

Example: Printer conflict (打印机冲突)

A process must request access to a device (e.g., printer)

The OS grants or denies access depending on availability

A flag indicates whether a resource is available

clear (0) → 可用

set (1) → 已占用

- Problem: The flag test and set may be interrupted

Two processes may access the resource simultaneously

race condition (竞争条件)

- Solution 1: Disable interrupts

Disable interrupts during the flag operation

- Solution 2: test-and-set

Test-and-set is a machine instruction that tests and sets a flag in one step.

It cannot be interrupted.

A **semaphore** is a properly implemented flag used to control access

critical region (临界区)

A critical region is a sequence of instructions that must be executed by only one process at a time

→ 临界区是一次只能被一个进程执行的代码段

mutual exclusion (互斥)

Mutual exclusion means only one process can access a critical region at a time

→ 互斥指同一时间只有一个进程可以进入临界区

Using semaphore

A process must set the semaphore before entering a critical region

→ 进入临界区前必须设置信号量

The process must clear the semaphore when leaving

→ 离开时必须清除信号量

If the semaphore is set, the process must wait

→ 如果信号量已被占用，进程必须等待

Deadlock

Deadlock (死锁)

Deadlock is a condition in which two or more processes are blocked because each is waiting for a resource held by another

→ 死锁是指两个或多个进程相互等待对方占有的资源，导致都无法继续执行

Forking (派生进程)

Forking is the creation of a new process by an existing process

→ Forking 是由已有进程创建新进程的操作

If all processes must create new processes but the process table is full, no process can continue

Deadlock Conditions

- Competition for nonsharable resources → 竞争不可共享资源
- Resources are requested on a partial basis → 资源是分阶段请求的
- Resources cannot be forcibly taken away → 资源一旦分配不能被强制收回

Deadlock Handling

- Deadlock detection and correction (检测与修复)

The system detects deadlock and forcibly retrieves resources

- Kill (终止进程)

Kill means forcibly terminating a process to release resources

→ Kill 指强制终止进程以释放资源

Deadlock avoidance

- Require processes to request all resources at once
- Convert nonsharable resources into sharable ones

Spooling

Spooling is a technique of storing output data in mass storage for later processing

→ Spooling 是将输出数据暂存到外存，稍后再处理

print_A.txt → 打印机

print_B.txt → 打印机

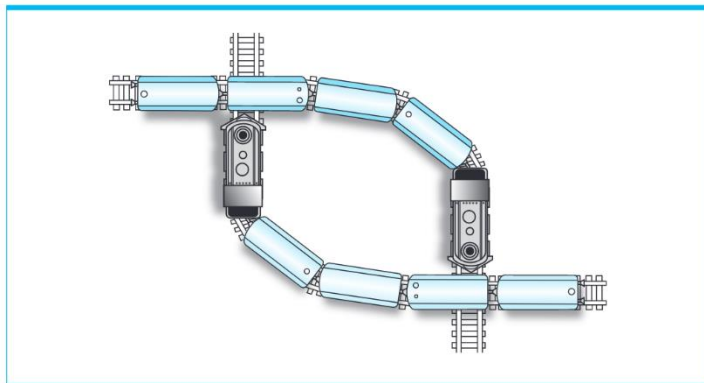
Python and Operating Systems

Programming languages like Python provide interfaces to interact with the operating system

Multi-Core Operating Systems

Multi-core systems use multiple processors to execute tasks simultaneously

Figure 3.7 A deadlock resulting from competition for nonsharable railroad intersections



1. Suppose process A and process B are sharing time on the same machine, and each needs the same nonsharable resource for short periods of time. (For example, each process may be printing a series of independent, short reports.) Each process may then repeatedly acquire the resource, release it, and later request it again. What is a drawback to controlling access to the resource in the following manner:

Begin by assigning a flag the value 0. If process A requests the resource and the flag is 0, grant the request. Otherwise, make process A wait. If process B requests the resource and the flag is 1, grant the request. Otherwise, make process B wait. Each time process A finishes with the resource, change the flag to 1. Each time process B finishes with the resource, change the flag to 0.

This method enforces strict alternation, which may cause a process to wait even when the resource is free, leading to inefficient use of the resource.

flag = 1 (轮到 B)

但 B 没请求资源

A 想用 → 却被阻塞 ❌

2. Suppose a two-lane road converges to one lane to pass through a tunnel. To coordinate the use of the tunnel, the following signal system has been installed:

A car entering either end of the tunnel causes red lights above the tunnel entrances to be turned on.

As the car exits the tunnel, the lights are turned off. If an approaching car finds a red light on, it waits until the light is turned off before entering the tunnel.

What is the flaw in this system?

The flaw is that two cars may enter the tunnel simultaneously if they both check the light before it is turned on.

The test and set of the signal are not atomic operations → 对信号的“检查和设置”不是原子操作

可能发生：

车 A 到达 → 看到绿灯 (还没变红)

在它进入前，被“打断” (类比 CPU 中断)

车 B 到达 → 也看到绿灯

两辆车同时进入隧道 ❌

3. Suppose the following solutions have been proposed for removing the deadlock that occurs on a single-lane bridge when two cars meet. Identify which condition for deadlock given in the text is removed by each solution.

a. Do not let a car onto the bridge until the bridge is empty.

This removes the condition of partial allocation of resources.

车必须“一次性获得整个桥” (不能占一半再等)

b. If cars meet, make one of them back up.

This removes the condition that resources cannot be forcibly taken away.

c. Add a second lane to the bridge.

This removes the competition for nonsharable resources.

4. Suppose we represent each process in a multiprocessing system with a dot and draw an arrow from one dot to another if the process represented by the first dot is waiting for a (nonsharable) resource being used by the second. Mathematicians call the resulting picture a **directed graph. What property of the directed graph is equivalent to deadlock in the system?**

A cycle in the directed graph indicates that a group of processes are waiting for each other in a circular manner, which results in deadlock.

A → B → C → A

A 等 B

B 等 C

C 等 A

3.5 Security

Security in an operating system includes protecting data and ensuring system reliability.

Attacks from the Outside

Login is the procedure by which a user establishes contact with the operating system.

super user/administrator has privileged access to the system

auditing software (审计软件) records and analyzes system activities

It can detect unusual login attempts or abnormal user behavior

sniffing software (嗅探软件) records user activities and reports them to intruders

It may simulate a login procedure to steal passwords

Attacks from Within (内部攻击)

Once access is gained, an intruder may explore the system or insert malicious software

Memory Protection

The CPU uses special registers to store memory limits of a process

Each memory access is checked against these limits

If the access is illegal, an interrupt occurs

Security Problem A process could try to modify the memory limit registers

privilege levels (特权级)

CPUs operate in two privilege levels: privileged mode and nonprivileged mode

Privileged mode allows execution of all instructions

Nonprivileged mode restricts certain instructions

privileged instructions

Privileged instructions can only be executed in privileged mode

Executing a privileged instruction in nonprivileged mode causes an interrupt

1. Give some examples of poor choices for passwords and explain why they would be poor choices.

Poor password choices include names, dates, and simple words because they are easy to guess.

2. Processors in Intel's Pentium series provide for four privilege levels. Why would the designers of CPUs decide to provide four levels rather than three or five?

Level	名称	作用
Ring 0	Kernel	操作系统内核
Ring 1	System services	系统服务
Ring 2	Drivers	设备驱动
Ring 3	User	用户程序

Four levels balance security and system complexity.

3. If a process in a multiprogramming system could access memory cells outside its allotted area, how could it gain control of the machine?

If a process could access memory outside its allotted area, it could modify or erase the operating system in memory.